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Reducing cavitation, noise, and vibration in a pressure exchanger

Abstract

A pressure exchanger includes a rotor configured to exchange pressure between a first fluid at a first pressure and a second fluid at a second pressure. The rotor forms ducts that are routed from a first distal end to a second distal end. The rotor forms chamfers on trailing edge rotor duct walls. The pressure exchanger further includes a first end cover that forms a high pressure in (HPIN) port configured to provide the first fluid at the first pressure into the ducts. The first end cover forms a low pressure out (LPOUT) port configured to receive the first fluid from the ducts at a third pressure. The pressure exchanger further includes a second end cover that forms a low pressure in (LPIN) port configured to provide the second fluid at the second pressure into the ducts and forms a high pressure out (HPOUT) port configured to receive the second fluid from the ducts at a fourth pressure.

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Background/Summary

RELATED APPLICATIONS (1) This application claims benefit of U.S. Provisional Application No. 63/395,694, filed Aug. 5, 2022, U.S. Provisional Application 63/388,991, filed Jul. 13, 2022, and U.S. Provisional Application 63/323,462, filed Mar. 24, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to pressure exchangers, and, more particularly, reducing cavitation, noise, and vibration in pressure exchangers.

BACKGROUND

(2) Pressure exchangers exchange pressure between a fluids. Mixing of the fluids may occur.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is illustrated by way of example, and not by way of limitation in the figures of the accompanying drawings.

(2) FIGS. 1A-D illustrate schematic diagrams of fluid handling systems including hydraulic energy transfer systems, according to certain embodiments.

(3) FIGS. 2A-E are exploded perspective views a pressure exchanger, according to certain embodiments.

(4) FIGS. 3A-P illustrate components of pressure exchangers, according to certain embodiments.

(5) FIGS. 4A-K illustrate components of pressure exchangers, according to certain embodiments.

(6) FIGS. 5A-J illustrate components of pressure exchangers, according to certain embodiments.

DETAILED DESCRIPTION OF EMBODIMENTS

(7) Embodiments described herein are related to optimizing pressure exchangers by one or more of minimizing mixing, enhancing efficiency, cavitation control, noise control, and/or vibration control.

(8) High pressure fluid may be used by systems, such as hydraulic fracturing (e.g., fracking or fracing) systems, desalinization systems, refrigeration systems, mud pumping systems, etc. Pumps may be used to provide the high pressure fluid. Certain fluids (e.g., brine, viscous fluids, sand, powder, debris, ceramics, etc.) may damage and reduce efficiency of pumps. A pressure exchanger may be used to exchange pressure between two fluids. A pump may be used to increase the pressure of a first fluid (e.g., substantially solid particle free, less viscous, water, etc.). A pressure exchanger may receive the high pressure first fluid (e.g., water) and a low pressure second fluid (e.g., fluid containing solid particles, fluid that is more viscous, brine) and may transfer the pressure from the high pressure first fluid to the low pressure second fluid.

(9) In a pressure exchanger, liquid-to-liquid pressure exchange make take place via an oscillating “fluid plug” in the rotor ducts. The “fluid plug” is not impermeable and mixing can occur between the two fluids that are exchanging pressure energy. Mixing may be dependent on multiple factors such as travel distance of the “fluid plug” (e.g., portion of the rotor duct that the “fluid plug” traverses, distance the “fluid plug” travels inside the rotor duct over one revolution normalized rotor length), turbulence, diffusion, jetting, rotor entry and exit losses, etc. Efficiency of a pressure exchanger may be directly proportional to the travel distance of the “fluid plug,” whereas pressure exchanger mixing may be inversely proportional to the travel distance. Conventional attempts to increase efficiency of the pressure exchanger by increasing travel distance also increase mixing

inside the pressure exchanger. Less efficient pressure exchangers use more energy and cause other components (e.g., pumps) to be used more and to be worn down more. Pressure exchangers that mix fluids cause increased specific energy consumption (e.g., in a sea water reverse osmosis (SWRO) plant), cause contamination of fluid (e.g., when performing pressure exchange with toxic fluid, fluid with solid particles, more viscous fluid, etc.), cause wearing down of components (e.g., wearing down pumps due to solid particles being introduced into the water), etc.

(10) In pressure exchangers (e.g., rotary isobaric pressure exchangers), rotating ducts carry high pressure (HP) fluids from HP ports (e.g., kidneys) to low pressure (LP) ports and also LP fluids from LP ports to HP ports. The duct fluids undergo rapid pressurization or depressurization when the duct fluids approach a set of ports. The frequency and the rate of pressurization or depressurization depends on the rotor revolutions per minute (RPM), number of pressure exchange cycles per revolution, and the pressure differential between HP and LP ports. The rapid pressurization and depressurization results in high velocity fluid jets generating noise and producing flow and pressure pulsations that increase vibration levels. This can also produce vapor bubbles if local fluid pressure (e.g., due to high velocity) falls below the vapor pressure at that temperature. The vapor bubbles collapse as the vapor bubbles travel to higher pressure regions which results in void formation which causes surrounding fluid to rush in producing very high localized pressure spikes. When occurring next to a solid wall, this produces pitting damage (e.g., cavitation), which accumulates over time. This cavitation further amplifies the noise and vibration levels.

(11) The devices, systems, and methods of the present disclosure provide optimizing of a pressure exchanger for efficiency, mixing, cavitation, sound, and vibration.

(12) A pressure exchanger includes a rotor configured to exchange pressure between a first fluid at a first pressure and a second fluid at a second pressure. The rotor forms ducts that are routed from a first distal end to a second distal end. The pressure exchanger further includes a first end cover that forms a high pressure in (HPIN) port configured to provide the first fluid at the first pressure into the ducts. The first end cover forms a low pressure out (LPOUT) port configured to receive the first fluid from the ducts at a third pressure. The pressure exchanger further includes a second end cover that forms a low pressure in (LPIN) port configured to provide the second fluid at the second pressure into the ducts and forms a high pressure out (HPOUT) port configured to receive the second fluid from the ducts at a fourth pressure.

(13) In some embodiments, the HPIN port is configured to provide the first fluid at the first pressure in a substantially axial direction into the ducts. The first end cover may include one or more of radial sidewalls that are closer to each other than radial sidewalls of the HPOUT port, a fillet between a radial sidewall and an inner sidewall, radial sidewalls that both form ramps, non-planar three-dimensional ramps, an insert, etc.

(14) In some embodiments, one or more sidewalls (e.g., radial sidewalls) of the first end cover that form the HPIN port or the LPOUT port are substantially planar (e.g., not curved). In some embodiments, the rotor forms ducts in at least three concentric rows. In some embodiments, the pressure exchanger includes spacers disposed in the first end cover, where the spacers are disposed between interconnects (e.g., metal interconnects) and the first end cover (e.g., ceramic end cover).

(15) In some embodiments, the second end cover forms a first spot face proximate the LPIN port and a second spot face proximate the HPOUT port. In some embodiments, the first end cover does not include spot faces **340** proximate the LPOUT port and the HPIN port. In some embodiments, the first and/or second spot faces **340** include a chamfer. In some embodiments, the first and/or second spot faces **340** include a radial extent that is shorter than a duct radial extent of a corresponding duct formed by the rotor. In some embodiments, the first and/or second spot faces **340** include different recesses for different concentric rows of the ducts. In some embodiments, the rotor forms ducts in concentric rows that are staggered. In some embodiments, the second end cover forms a pre-pressurization hole proximate the first spot face.

(16) The present disclosure has advantages over conventional solutions. In some embodiments, pressure exchangers of the present disclosure exchange pressure at higher efficiencies compared to conventional systems. This provides for less energy usage and less use and wear of other components (e.g., pumps). In some embodiments, pressure exchangers of the present disclosure exchange pressure with less mixing of fluids compared to conventional systems. This provides for less specific energy consumption of plants, less contamination of fluids, less wearing down of components (e.g., pumps), etc. In some embodiments, pressure exchangers of the present disclosure generate less noise, have less vibration, and have less damage (e.g., pitting damage, cavitation) compared to conventional systems. This provides for less wear, less maintenance, less replacement of components, less downtime, etc.

(17) Although some embodiments of the present disclosure are described in relation to the rotor and end covers of pressure exchangers, embodiments of the current disclosure can be applied to other components and other devices (e.g., adapter plates, etc.).

(18) Although some embodiments of the present disclosure are described in relation to isobaric pressure exchangers, pressure exchangers, and hydraulic energy transfer systems, the current disclosure can be applied to other systems and devices (e.g., pressure exchanger that is not isobaric, rotating components that are not a pressure exchanger, a pressure exchanger that is not rotary, etc.).

(19) Although some embodiments of the present disclosure are described in relation to exchanging pressure between fluid used in fracing systems, desalinization systems, and/or refrigeration systems, the present disclosure can be applied to other types of systems. Fluids can refer to liquid, gas, transcritical fluid, supercritical fluid, subcritical fluid, and/or combinations thereof.

(20) Although some embodiments of the present disclosure are described in relation to a pressure exchanger that has a sleeve, in some embodiments, the pressure exchanger of the present disclosure is sleeved (e.g., has a sleeve) and in some embodiments, the pressure exchanger of the present disclosure is sleeve-less (e.g., does not have a sleeve, has a center-post).

(21) Although some embodiments of the present disclosure are described in relation to a pressure exchanger that has a single cycle, in some embodiments, the pressure exchanger of the present disclosure is a multi-cycle pressure exchanger.

(22) FIGS. 1A-D illustrate schematic diagrams of fluid handling systems **100A-D** including hydraulic energy transfer systems **110** (e.g., pressure exchangers), according to certain embodiments.

(23) Each hydraulic energy transfer system **110** of FIGS. 1A-D may include a pressure exchanger that includes a rotor configured to exchange pressure between a first fluid at a first pressure and a second fluid at a second pressure. The rotor forms ducts that are routed from a first distal end to a second distal end. The pressure exchanger further includes a first end cover that forms a high pressure in (HPIN) port configured to provide the first fluid at the first pressure into the ducts. The first end cover forms a low pressure out (LPOUT) port configured to receive the first fluid from the ducts at a third pressure. The pressure exchanger further includes a second end cover that forms a low pressure in (LPIN) port configured to provide the second fluid at the second pressure into the ducts and forms a high pressure out (HPOUT) port configured to receive the second fluid from the ducts at a fourth pressure.

(24) In some embodiments, the hydraulic energy transfer system **110** of FIGS. 1A-D has a HPIN port that is configured to provide the first fluid at the first pressure in a substantially axial direction into the ducts of the rotor of the hydraulic energy transfer system **110**. In some embodiments, one or more sidewalls (e.g., radial sidewalls) of the end cover that form the HPIN port or the LPOUT port are substantially planar (e.g., not curved). In some embodiments, the second end cover forms a first spot face proximate the LPIN port and a second spot face proximate the HPOUT port.

(25) FIG. 1A illustrates a schematic diagram of a fluid handling system **100A** including a hydraulic energy transfer system **110**, according to certain embodiments.

(26) In some embodiments, a hydraulic energy transfer system **110** includes a pressure exchanger

(PX). The hydraulic energy transfer system **110** (e.g., PX) receives low pressure (LP) fluid in **120** (e.g., low-pressure inlet stream, low pressure in (LPIN)) from a LP in system **122**. The hydraulic energy transfer system **110** also receives high pressure (HP) fluid in **130** (e.g., high-pressure inlet stream, high pressure in (HPIN)) from HP in system **132**. The hydraulic energy transfer system **110** (e.g., PX) exchanges pressure between the HP fluid in **130** and the LP fluid in **120** to provide LP fluid out **140** (e.g., low-pressure outlet stream, low pressure out (LPOUT)) to LP fluid out system **142** and to provide HP fluid out **150** (e.g., high-pressure outlet stream, high pressure out (HPOUT)) to HP fluid out system **152**.

(27) In some embodiments, the hydraulic energy transfer system **110** includes a PX to exchange pressure between the HP fluid in **130** and the LP fluid in **120**. The PX may be referred to as an isobaric pressure exchanger (IPX). The PX (e.g., an IPX) may be a device that transfers fluid pressure between HP fluid in **130** and LP fluid in **120** at efficiencies in excess of approximately 50%, 60%, 70%, 80%, 90%, or greater (e.g., without utilizing centrifugal technology, substantially isobaric pressure exchange). High pressure (e.g., HP fluid in **130**, HP fluid out **150**) refers to pressures greater than the low pressure (e.g., LP fluid in **120**, LP fluid out **140**). LP fluid in **120** of the PX may be pressurized and exit the PX at high pressure (e.g., HP fluid out **150**, at a pressure greater than that of LP fluid in **120**), and HP fluid in **130** may be depressurized and exit the PX at low pressure (e.g., LP fluid out **140**, at a pressure less than that of the HP fluid in **130**). The PX may operate with the HP fluid in **130** directly applying a force to pressurize the LP fluid in **120**, with or without a fluid separator between the fluids. Examples of fluid separators that may be used with the PX include, but are not limited to, pistons, bladders, diaphragms and the like. In some embodiments, PXs may be rotary devices. Rotary PXs, such as those manufactured by Energy Recovery, Inc. of San Leandro, Calif, may not have any separate valves, since the effective valving action is accomplished internal to the device via the relative motion of a rotor with respect to end covers. Rotary PXs may be designed to operate with internal pistons to isolate fluids and transfer pressure with relatively little mixing of the inlet fluid streams. Reciprocating PXs may include a piston moving back and forth in a cylinder for transferring pressure between the fluid streams. Any PX or multiple PXs may be used in the present disclosure, such as, but not limited to, rotary PXs, reciprocating PXs, or any combination thereof. In addition, the PX may be disposed on a skid separate from the other components of a fluid handling system **100** (e.g., in situations in which the PX is added to an existing fluid handling system).

(28) In some embodiments, a motor **160** is coupled to hydraulic energy transfer system **110** (e.g., to a PX). In some embodiments, the motor **160** controls the speed (e.g., RPM) of a rotor of the hydraulic energy transfer system **110** (e.g., to increase pressure of HP fluid out **150**, to decrease pressure of LP fluid out **140**, etc.). In some embodiments, motor **160** generates energy (e.g., acts as a generator) based on pressure exchanging in hydraulic energy transfer system **110**.

(29) The hydraulic energy transfer system **110** may be a hydraulic protection system (e.g., hydraulic buffer system, hydraulic isolation system) that may block or limit contact between fluid (e.g., solid particle laden fluid, frac fluid, viscous fluid, toxic fluid, etc.) and various equipment (e.g., hydraulic fracturing equipment, high-pressure pumps) while exchanging work and/or pressure with another fluid. By blocking or limiting contact between various equipment (e.g., fracturing equipment, pumps, etc.) and particular fluids (e.g., solid particle containing fluid), the hydraulic energy transfer system **110** increases the life and performance, while reducing abrasion, wear, contamination, etc., of various equipment (e.g., fracturing equipment, high pressure fluid pumps). Less expensive equipment may be used in the fluid handling system **100** by using equipment (e.g., high pressure fluid pumps) not designed for abrasive fluids (e.g., frac fluids and/or corrosive fluids).

(30) The hydraulic energy transfer system **110** may include a hydraulic turbocharger or hydraulic pressure exchange system, such as a rotating PX. The PX may include one or more chambers (e.g., 1 to 100) to facilitate pressure transfer and equalization of pressures between volumes of first and

second fluids (e.g., gas, liquid, multi-phase fluid). In some embodiments, the PX may transfer pressure between a first fluid (e.g., pressure exchange fluid, such as a proppant free or substantially proppant free fluid) and a second fluid that may be highly viscous and/or contain solid particles (e.g., frac fluid containing sand, proppant, powders, debris, ceramics). The solid particle fluid causes abrasion and/or erosion of components of the PX, such as the rotor and end covers of the PX. The fluid (e.g., abrasive particles in the fluid) may cause wear to an interface between the rotor and each end cover as the rotor rotates relative to the end covers. Replacing worn components of the PX may be costly.

(31) The hydraulic energy transfer system **110** may be used in different types of systems, such as fracing systems, desalination systems, refrigeration systems, etc.

(32) FIG. **1B** illustrates a schematic diagram of a fluid handling system **100B** including a hydraulic energy transfer system **110**, according to certain embodiments. Features in FIG. **1B** that have the same or similar reference numbers as features in FIG. **1A** may have the same or similar components, features, etc. as those in FIG. **1A**.

(33) Fluid handling system **100B** may be a fracing system (e.g., hydraulic fracturing system). In some embodiments, fluid handling system **100B** includes more components, less components, same routing, different routing, and/or the like than that shown in FIG. **1B**.

(34) LP fluid in **120** and HP fluid out **150** may be frac fluid (e.g., fluid including solid particles, proppant fluid, etc.). HP fluid in **130** and LP fluid out **140** may be substantially solid particle free fluid (e.g., proppant free fluid, water, filtered fluid, etc.).

(35) LP in system **122** may include one or more low pressure fluid pumps to provide LP fluid in **120** to the hydraulic energy transfer system **110** (e.g., PX). HP in system **132** may include one or more high pressure fluid pumps **134** to provide HP fluid in **130** to hydraulic energy transfer system **110**.

(36) Hydraulic energy transfer system **110** exchanges pressure between LP fluid in **120** (e.g., low pressure frac fluid) and HP fluid in **130** (e.g., high pressure water) to provide HP fluid out **150** (e.g., high pressure frac fluid) to HP out system **152** and to provide LP fluid out **140** (e.g., low pressure water). HP out system **152** may include a rock formation **154** (e.g., well) that includes cracks **156**. The solid particles (e.g., proppants) from HP fluid out **150** may be provided into the cracks **156** of the rock formation.

(37) In some embodiments, LP fluid out **140**, high pressure fluid pumps **134**, and HP fluid in **130** are part of a first loop (e.g., proppant free fluid loop). The LP fluid out **140** may be provided to the high pressure fluid pumps to generate HP fluid in **130** that becomes LP fluid out **140** upon exiting the hydraulic energy transfer system **110**.

(38) In some embodiments, LP fluid in **120**, HP fluid out **150**, and low pressure fluid pumps **124** are part of a second loop (e.g., proppant containing fluid loop). The HP fluid out **150** may be provided into the rock formation **154** and then pumped from the rock formation **154** by the low pressure fluid pumps **124** to generate LP fluid in **120**.

(39) In some embodiments, fluid handling system **100B** is used in well completion operations in the oil and gas industry to perform hydraulic fracturing (e.g., fracking, fracing) to increase the release of oil and gas in rock formations **154**. HP out system **152** may include rock formations **154** (e.g., a well). Hydraulic fracturing may include pumping HP fluid out **150** containing a combination of water, chemicals, and solid particles (e.g., sand, ceramics, proppant) into a well (e.g., rock formation **154**) at high pressures. LP fluid in **120** and HP fluid out **150** may include a particulate laden fluid that increases the release of oil and gas in rock formations **154** by propagating and increasing the size of cracks **156** in the rock formations **154**. The high pressures of HP fluid out **150** initiates and increases size of cracks **156** and propagation through the rock formation **154** to release more oil and gas, while the solid particles (e.g., powders, debris, etc.) enter the cracks **156** to keep the cracks **156** open (e.g., prevent the cracks **156** from closing once HP fluid out **150** is depressurized).

(40) In order to pump this particulate laden fluid into the rock formation **154** (e.g., a well), the fluid handling system **100B** may include one or more high pressure fluid pumps **134** and one or more low pressure fluid pumps **124** coupled to the hydraulic energy transfer system **110**. For example, the hydraulic energy transfer system **110** may be a hydraulic turbocharger or a PX (e.g., a rotary PX). In operation, the hydraulic energy transfer system **110** transfers pressures without any substantial mixing between a first fluid (e.g., HP fluid in **130**, proppant free fluid) pumped by the high pressure fluid pumps **134** and a second fluid (e.g., LP fluid in **120**, proppant containing fluid, frac fluid) pumped by the low pressure fluid pumps **124**. In this manner, the hydraulic energy transfer system **110** blocks or limits wear on the high pressure fluid pumps **134**, while enabling the fluid handling system **100B** to pump a high-pressure frac fluid (e.g., HP fluid out **150**) into the rock formation **154** to release oil and gas. In order to operate in corrosive and abrasive environments, the hydraulic energy transfer system **110** may be made from materials resistant to corrosive and abrasive substances in either the first and second fluids. For example, the hydraulic energy transfer system **110** may be made out of ceramics alumina, cermets, such as carbide, oxide, nitride, or boride hard phases) within a metal matrix (e.g., Co, Cr or Ni or any combination thereof) such as tungsten carbide in a matrix of CoCr, Ni, NiCr or Co.

(41) In some embodiments, the hydraulic energy transfer system **110** includes a PX (e.g., rotary PX) and HP fluid in **130** (e.g., the first fluid, high-pressure solid particle free fluid) enters a first side of the PX where the HP fluid in **130** contacts LP fluid in **120** (e.g., the second fluid, low-pressure frac fluid) entering the PX on a second side. The contact between the fluids enables the HP fluid in **130** to increase the pressure of the second fluid (e.g., LP fluid in **120**), which drives the second fluid out (e.g., HP fluid out **150**) of the PX and down a well (e.g., rock formation **154**) for fracturing operations. The first fluid (e.g., LP fluid out **140**) similarly exits the PX, but at a low pressure after exchanging pressure with the second fluid. As noted above, the second fluid may be a low-pressure frac fluid that may include abrasive particles, which may wear the interface between the rotor and the respective end covers as the rotor rotates relative to the respective end covers.

(42) FIG. 1C illustrates a schematic diagram of a fluid handling system **100C** including a hydraulic energy transfer system **110**, according to certain embodiments. Fluid handling system **100C** may be a desalination system (e.g., remove salt and/or other minerals from water). In some embodiments, fluid handling system **100C** includes more components, less components, same routing, different routing, and/or the like than that shown in FIG. 1C.

(43) LP in system **122** may include a feed pump **126** (e.g., low pressure fluid pump **124**) that receives seawater in **170** (e.g., feed water from a reservoir or directly from the ocean) and provides LP fluid in **120** (e.g., low pressure seawater, feed water) to hydraulic energy transfer system **110** (e.g., PX). HP in system **132** may include membranes **136** that provide HP fluid in **130** (e.g., high pressure brine) to hydraulic energy transfer system **110** (e.g., PX). The hydraulic energy transfer system **110** exchanges pressure between the HP fluid in **130** and LP fluid in **120** to provide HP fluid out **150** (e.g., high pressure seawater) to HP out system **152** and to provide LP fluid out **140** (e.g., low pressure brine) to LP out system **142** (e.g., geological mass, ocean, sea, discarded, etc.).

(44) The membranes **136** may be a membrane separation device configured to separate fluids traversing a membrane, such as a reverse osmosis membrane. Membranes **136** may provide HP fluid in **130** which is a concentrated feed-water or concentrate (e.g., brine) to the hydraulic energy transfer system **110**. Pressure of the HP fluid in **130** may be used to compress low-pressure feed water (e.g., LP fluid in **120**) to be high pressure feed water (e.g., HP fluid out **150**). For simplicity and illustration purposes, the term feed water is used. However, fluids other than water may be used in the hydraulic energy transfer system **110**. The hydraulic energy transfer system **110** can also be used for other applications, such as industrial waste water.

(45) The circulation pump **158** (e.g., centrifugal pump) provides the HP fluid out **150** (e.g., high pressure seawater) to membranes **136**. The membranes **136** filter the HP fluid out **150** to provide LP potable water **172** and HP fluid in **130** (e.g., high pressure brine). The LP out system **142**

provides brine out **174** (e.g., to geological mass, ocean, sea, discarded, etc.).

(46) In some embodiments, a high pressure fluid pump **176** is disposed between the feed pump **126** and the membranes **136**. The high pressure fluid pump **176** increases pressure of the low pressure seawater (e.g., LP fluid in **120**, provides high pressure feed water) to be mixed with the high pressure seawater provided by circulation pump **158**.

(47) In some embodiments, use of the hydraulic energy transfer system **110** decreases the load on high pressure fluid pump **176**. In some embodiments, fluid handling system **100C** provides LP potable water **172** without use of high pressure fluid pump **176**. In some embodiments, fluid handling system **100C** provides LP potable water **172** with intermittent use of high pressure fluid pump **176**.

(48) In some examples, hydraulic energy transfer system **110** (e.g., PX) receives LP fluid in **120** (e.g., low-pressure feed-water) at about 30 pounds per square inch (PSI) and receives HP fluid in **130** (e.g., high-pressure brine or concentrate) at about 980 PSI. The hydraulic energy transfer system **110** (e.g., PX) transfers pressure from the high-pressure concentrate (e.g., HP fluid in **130**) to the low-pressure feed-water (e.g., LP fluid in **120**). The hydraulic energy transfer system **110** (e.g., PX) outputs HP fluid out **150** (e.g., high pressure (compressed) feed-water) at about 965 PSI and LP fluid out **140** (e.g., low-pressure concentrate) at about 15 PSI. Thus, the hydraulic energy transfer system **110** (e.g., PX) may be about 97% efficient since the input volume is about equal to the output volume of the hydraulic energy transfer system **110** (e.g., PX), and 965 PSI is about 97% of 980 PSI.

(49) FIG. 1D illustrates a schematic diagram of a fluid handling system **100D** including a hydraulic energy transfer system **110**, according to certain embodiments. Fluid handling system **100D** may be a refrigeration system. In some embodiments, fluid handling system **100D** includes more components, less components, same routing, different routing, and/or the like than that shown in FIG. 1D.

(50) Hydraulic energy transfer system **110** (e.g., PX) may receive LP fluid in **120** from LP in system **122** (e.g., low pressure lift device **128**, low pressure fluid pump, etc.) and HP fluid in **130** from HP in system **132** (e.g., condenser **138**). The hydraulic energy transfer system **110** (e.g., PX) may exchange pressure between the LP fluid in **120** and HP fluid in **130** to provide HP fluid out **150** to HP out system **152** (e.g., high pressure lift device **159**) and to provide LP fluid out **140** to LP out system **142** (e.g., evaporator **144**). The evaporator **144** may provide the fluid to compressor **178** and low pressure lift device **128**. The condenser **138** may receive fluid from compressor **178** and high pressure lift device **159**.

(51) The fluid handling system **100D** may be a closed system. LP fluid in **120**, HP fluid in **130**, LP fluid out **140**, and HP fluid out **150** may all be a fluid (e.g., refrigerant) that is circulated in the closed system of fluid handling system **100D**.

(52) In some embodiments, the fluid of fluid handling system **100D** may include solid particles. For example, the piping, equipment, connections (e.g., pipe welds, pipe soldering), etc. may introduce solid particles (e.g., solid particles from the welds) into the fluid in the fluid handling system **100D**. The solid particles in the fluid and/or the high pressure of the fluid may cause abrasion and/or erosion of components (e.g., rotor, end covers) of the PX of hydraulic energy transfer system **110**.

(53) FIGS. 2A-E are exploded perspective views a rotary PX **40** (e.g., rotary pressure exchanger, rotary liquid piston compressor (LPC)), according to certain embodiments.

(54) PX **40** includes a rotor **46** configured to exchange pressure between a first fluid at a first pressure and a second fluid at a second pressure. The rotor **46** forms channels **70** (e.g., ducts) that are routed from a first distal end (e.g., opening **72**) to a second distal end (e.g., opening **74**). The PX **40** further includes an end cover **64** that forms a HPIN port (e.g., outlet aperture **76**) configured to provide the first fluid at the first pressure into the channels **70**. The end cover **64** forms a LPOUT port (e.g., outlet aperture **78**) configured to receive the first fluid from the channels **70** at a third

pressure. The PX **40** further includes an end cover **66** that forms a LPIN port (e.g., outlet aperture **80**) configured to provide the second fluid at the second pressure into the channels **70** and forms a HPOUT port (e.g., outlet aperture **82**) configured to receive the second fluid from the channels **70** at a fourth pressure.

(55) In some embodiments, the PX **40** of FIGS. 2A-E has a HPIN port (e.g., outlet aperture **76**) that is configured to provide the first fluid at the first pressure in a substantially axial direction into the channels **70** of the rotor **46** of the PX **40**. In some embodiments, one or more sidewalls (e.g., radial sidewalls) of the end cover **64** that form the HPIN port (e.g., outlet aperture **76**) or the LPOUT port are substantially planar (e.g., not curved). In some embodiments, the end cover **66** forms a first spot face proximate the LPIN port (e.g., outlet aperture **80**) and a second spot face proximate the HPOUT port (e.g., outlet aperture **82**).

(56) PX **40** is configured to transfer pressure and/or work between a first fluid (e.g., proppant free fluid or supercritical carbon dioxide, HP fluid in **130**) and a second fluid (e.g., frac fluid or superheated gaseous carbon dioxide, LP fluid in **120**) with minimal mixing of the fluids. The rotary PX **40** may include a generally cylindrical body portion **42** that includes a sleeve **44** (e.g., rotor sleeve) and a rotor **46**. The rotary PX **40** may also include two end caps **48** and **50** that include manifolds **52** and **54**, respectively. Manifold **52** includes respective inlet port **56** and outlet port **58**, while manifold **54** includes respective inlet port **60** and outlet port **62**. In operation, these inlet ports **56**, **60** enable the first and second fluids to enter the rotary PX **40** to exchange pressure, while the outlet ports **58**, **62** enable the first and second fluids to then exit the rotary PX **40**. In operation, the inlet port **56** may receive a high-pressure first fluid (e.g., HP fluid in **130**), and after exchanging pressure, the outlet port **58** may be used to route a low-pressure first fluid (e.g., LP fluid out **140**) out of the rotary PX **40**. Similarly, the inlet port **60** may receive a low-pressure second fluid (e.g., LP fluid in **120**) and the outlet port **62** may be used to route a high-pressure second fluid (e.g., HP fluid out **150**) out of the rotary PX **40**. The end caps **48** and **50** include respective end covers **64** and **66** (e.g., end plates) disposed within respective manifolds **52** and **54** that enable fluid sealing contact with the rotor **46**.

(57) As noted above, one or more components of the PX **40**, such as the rotor **46**, the end cover **64**, and/or the end cover **66**, may be constructed from a wear-resistant material (e.g., carbide, cemented carbide, silicon carbide, tungsten carbide, etc.) with a hardness greater than a predetermined threshold a Vickers hardness number that is at least 1000, 1250, 1500, 1750, 2000, 2250, or more). For example, tungsten carbide may be more durable and may provide improved wear resistance to abrasive fluids as compared to other materials, such as alumina ceramics.

(58) The rotor **46** may be cylindrical and disposed in the sleeve **44**, which enables the rotor **46** to rotate about the axis **68**. The rotor **46** may have a plurality of channels **70** (e.g., ducts, rotor ducts) extending substantially longitudinally through the rotor **46** with openings **72** and **74** (e.g., rotor ports) at each end arranged symmetrically about the longitudinal axis **68**. The openings **72** and **74** of the rotor **46** are arranged for hydraulic communication with inlet and outlet apertures **76** and **78** (e.g., end cover inlet port and end cover outlet port) and **80** and **82** (e.g., end cover inlet port and end cover outlet port) in the end covers **64** and **66**, in such a manner that during rotation the channels **70** are exposed to fluid at high-pressure and fluid at low-pressure. As illustrated, the inlet and outlet apertures **76** and **78** and **80** and **82** may be designed in the form of arcs or segments of a circle (e.g., C-shaped).

(59) In some embodiments, a controller using sensor feedback (e.g., revolutions per minute measured through a tachometer or optical encoder or volume flow rate measured through flowmeter) may control the extent of mixing between the first and second fluids in the rotary PX **40**, which may be used to improve the operability of the fluid handling system (e.g., fluid handling systems **100A-D** of FIGS. 1A-D). For example, varying the volume flow rates of the first and second fluids entering the rotary PX **40** allows the plant operator (e.g., system operator) to control the amount of fluid mixing within the PX **40**. In addition, varying the rotational speed of the rotor

46 also allows the operator to control mixing. Three characteristics of the rotary PX **40** that affect mixing are: (1) the aspect ratio of the rotor channels **70**; (2) the duration of exposure between the first and second fluids; and (3) the creation of a fluid barrier (e.g., an interface) between the first and second fluids within the rotor channels **70**. First, the rotor channels **70** (e.g., ducts) are generally long and narrow, which stabilizes the flow within the rotary PX **40**. In addition, the first and second fluids may move through the channels **70** in a plug flow regime with minimal axial mixing. Second, in certain embodiments, the speed of the rotor **46** reduces contact between the first and second fluids. For example, the speed of the rotor **46** (e.g., rotor speed of approximately 1200 RPM) may reduce contact times between the first and second fluids to less than approximately 0.15 seconds, 0.10 seconds, or 0.05 seconds. Third, a small portion of the rotor channel **70** is used for the exchange of pressure between the first and second fluids. Therefore, a volume of fluid remains in the channel **70** as a barrier between the first and second fluids. All these mechanisms may limit mixing within the rotary PX **40**. Moreover, in some embodiments, the rotary PX **40** may be designed to operate with internal pistons or other barriers, either complete or partial, that isolate the first and second fluids while enabling pressure transfer.

(60) FIGS. 2B-2E are exploded views of an embodiment of the rotary PX **40** illustrating the sequence of positions of a single rotor channel **70** in the rotor **46** as the channel **70** rotates through a complete cycle. It is noted that FIGS. 2B-2E are simplifications of the rotary PX **40** showing one rotor channel **70**, and the channel **70** is shown as having a circular cross-sectional shape. In other embodiments, the rotary PX **40** may include a plurality of channels **70** with the same or different cross-sectional shapes (e.g., circular, oval, square, rectangular, polygonal, etc.). Thus, FIGS. 2B-2E are simplifications for purposes of illustration, and other embodiments of the rotary PX **40** may have configurations different from that shown in FIGS. 2A-2E. As described in detail below, the rotary PX **40** facilitates pressure exchange between first and second fluids by enabling the first and second fluids to briefly contact each other within the rotor **46**. In certain embodiments, this exchange happens at speeds that result in limited mixing of the first and second fluids. The speed of the pressure wave traveling through the rotor channel **70** (as soon as the channel is exposed to the aperture **76**), the diffusion speeds of the fluids, and the rotational speed of rotor **46** dictate whether any mixing occurs and to what extent.

(61) FIG. 2B is an exploded perspective view of an embodiment of a rotary PX **40** (e.g., rotary LPC), according to certain embodiments. In FIG. 2B, the channel opening **72** is in a first position. In the first position, the channel opening **72** is in fluid communication with the aperture **78** in end cover **64** and therefore with the manifold **52**, while the opposing channel opening **74** is in hydraulic communication with the aperture **82** in end cover **66** and by extension with the manifold **54**. As will be discussed below, the rotor **46** may rotate in the clockwise direction indicated by arrow **84**. In operation, low-pressure second fluid **86** passes through end cover **66** and enters the channel **70**, where it contacts the first fluid **88** at a dynamic fluid interface **90**. The second fluid **86** then drives the first fluid **88** out of the channel **70**, through end cover **64**, and out of the rotary PX **40**. However, because of the short duration of contact, there is minimal mixing between the second fluid **86** and the first fluid **88**.

(62) FIG. 2C is an exploded perspective view of an embodiment of a rotary PX **40** (e.g., rotary LPC), according to certain embodiments. In FIG. 2C, the channel **70** has rotated clockwise through an arc of approximately 90 degrees. In this position, the opening **74** (e.g., outlet) is no longer in fluid communication with the apertures **80** and **82** of end cover **66**, and the opening **72** is no longer in fluid communication with the apertures **76** and **78** of end cover **64**. Accordingly, the low-pressure second fluid **86** is temporarily contained within the channel **70**.

(63) FIG. 2D is an exploded perspective view of an embodiment of a rotary PX **40** (e.g., rotary LPC), according to certain embodiments. In FIG. 2D, the channel **70** has rotated through approximately 60 degrees of arc from the position shown in FIG. 2B. The opening **74** is now in fluid communication with aperture **80** in end cover **66**, and the opening **72** of the channel **70** is now

in fluid communication with aperture **76** of the end cover **64**. In this position, high-pressure first fluid **88** enters and pressurizes the low-pressure second fluid **86**, driving the second fluid **86** out of the rotor channel **70** and through the aperture **80**.

(64) FIG. **2E** is an exploded perspective view of an embodiment of a rotary PX **40** (e.g., rotary LPC), according to certain embodiments. In FIG. **2E**, the channel **70** has rotated through approximately 270 degrees of arc from the position shown in FIG. **2B**. In this position, the opening **74** is no longer in fluid communication with the apertures **80** and **82** of end cover **66**, and the opening **72** is no longer in fluid communication with the apertures **76** and **78** of end cover **64**. Accordingly, the first fluid **88** is no longer pressurized and is temporarily contained within the channel **70** until the rotor **46** rotates another 90 degrees, starting the cycle over again.

(65) FIGS. **3A-P** illustrate components of pressure exchangers **300** (e.g., PXs **40** of FIGS. **2A-E**), according to certain embodiments. Features in FIGS. **3A-P** that have similar names and/or reference numbers as features in one or more of FIGS. **1A-2E** may include the same or similar structure, material, functionality, and/or the like as features in one or more of FIGS. **1A-2E**. In some embodiments, one or more features of PX **300** of FIGS. **3A-P** reduce mixing of fluids in the PX **300**.

(66) FIG. **3A** illustrates a perspective view of an end cover **310** (e.g., end cover **64** and/or **66** of one or more of FIGS. **2A-E**). FIG. **3B** illustrates a perspective cut-away view of an end cover **310** (e.g., of FIG. **3A**, end cover **64** and/or end cover **66** of one or more of FIGS. **2A-E**). FIG. **3C** illustrates a cross-sectional view of components of the pressure exchanger **300** (e.g., of FIGS. **3A** and/or **3B**, end cover **64** and/or **66** of one or more of FIGS. **2A-E**, rotor **46** of one or more of FIGS. **2A-E**).

(67) In some embodiments, a PX **300** includes a rotor **320** and end covers **310**. The rotor **320** is configured to rotate to exchange pressure between a first fluid at a first pressure and a second fluid at a second pressure. The rotor **320** forms ducts **322** (e.g., channel **70** of FIGS. **2A-E**) that are routed from a first distal end of the rotor **320** to a second distal end of the rotor **320**.

(68) The PX **300** may include first and second end covers **310**. The first end cover **310** is disposed at the first distal end of the rotor **320** and the second end cover **310** is disposed at the second distal end of the rotor **320**.

(69) End cover **310** is disposed at a distal end of the rotor **320**. End cover **310** forms ports **312** (e.g., apertures **76** and **78** of FIGS. **2A-E**, apertures **80** and **82** of FIGS. **2A-E**, inlet, outlet, HPIN port, HPOUT port, LP IN port, LPOUT port)

(70) In some embodiments, end cover **310** (e.g., disposed at a first distal end of the rotor **320**) forms an HPIN port configured to provide the first fluid at the first pressure in a substantially axial direction into the ducts **322** and an LPOUT port configured to receive the first fluid from the ducts **322** at a third pressure that is lower than the first pressure.

(71) In some embodiments, end cover **310** (e.g., disposed at a second distal end of the rotor **320**) forms an LPIN port configured to provide the second fluid at the second pressure into the ducts **322** and forms an HPOUT port configured to receive the second fluid from the ducts **322** at a fourth pressure that is higher than the second pressure.

(72) In some embodiments, PX **300** may include two ramps to reduced rotor incidence losses and to prevent skewing of the interface.

(73) In some embodiments, end cover insert **330** (e.g., made of polycarbonate, fiber reinforced polytetrafluoroethylene (PTFE) and/or polyetheretherketone (PEEK)) to guide the flow.

(74) In some embodiments, end cover **310** has 3D ramps.

(75) In some embodiments, end cover **310** has an HPIN kidney closure earlier than HPOUT by at least 2 degrees to avoid acceleration of flow as duct closes.

(76) In some embodiments, end cover **310** has a fillet (e.g., large fillet) at HPIN kidney exit diameter corner (e.g., to reduce mixing through inner row of ducts).

(77) In some embodiments, PX **300** has HPIN kidney opening later than HPOUT to help move interface away from HPOUT and to reduce mixing at HPOUT.

- (78) In some embodiments, spot faces are located at HPOUT open (and not on HPIN open) to reduce mixing at HPOUT and to avoid disturbing the interface which is close to HPIN open.
- (79) In some embodiments, HPOUT fluid is used as bearing fluid and center bore fluid (e.g., to reduce mixing of HPIN to HPOUT through leakage in axial gaps).
- (80) In some embodiments, length to depth (L/D) ratio and roundness of ducts **322** are maximized.
- (81) In some embodiments, honeycomb flow straighteners are used as inserts in the ducts **322** to reduce mixing.
- (82) In some embodiments, an end cover **310** includes radial sidewalls **314** (e.g., leading radial sidewall **314A**, trailing radial sidewall **314B**), an inner sidewall **316**, and an outer sidewall **318** that form a port **312** (e.g., HPIN port, HPOUT port, LPIN port, LPOUT). Leading radial sidewall **314A** is the first radial sidewall that a duct **322** sees as it rotates. The trailing radial sidewall **314B** is the sidewall of the same port that the rotor duct **322** sees after the leading radial side wall. In some embodiments, ports **312** have similar sizes. In some embodiments, the radial sidewalls **314** of the HPIN port are at least two degrees (e.g., four degrees, six degrees, eight degrees, ten degrees, etc.) closer to each other than radial sidewalls **314** of the HPOUT port are to each other (e.g., see FIGS. 3G-H). For example, FIG. 3G may be an HPOUT port and FIG. 3H may be an HPIN port.
- (83) In some embodiments, at least one of the radial sidewalls **314** (e.g., trailing radial sidewall) that form the HPIN port is configured to close relative to a duct **322** formed by the rotor **320** at least two degrees (e.g., four degrees, six degrees, eight degrees, ten degrees, etc.) prior to at least one of the second radial sidewalls (e.g., trailing radial sidewall) that form the HPOUT port closes relative to the duct **322** (e.g., to reduce mixing of corresponding fluid between HPIN and HPOUT, reduce mixing of HPIN into HPOUT).
- (84) In some embodiments, at least one of the radial sidewalls **314** (e.g., trailing radial sidewall) that form the LPIN port is configured to close relative to a duct **322** formed by the rotor **320** at least two degrees (e.g., four degrees, six degrees, eight degrees, ten degrees, etc.) prior to at least one of the second radial sidewalls (e.g., trailing radial sidewall) that form the LPOUT port closes relative to the duct **322** (e.g., to reduce mixing of corresponding fluid between LPIN and LPOUT, reduce mixing of LPIN into LPOUT).
- (85) In some embodiments, one or more radial sidewalls of the LPOUT port are biased to reduce mixing at the LPOUT port.
- (86) In some embodiments, each of the radial sidewalls **314** that form the HPIN port have a substantially straight edge that substantially matches a corresponding substantially straight edge of a duct **322** formed by the rotor **320** (e.g., to maximize the time of duct **322** exposure to the port to improve efficiency).
- (87) In some embodiments, a first end cover **310** forms a first port and a second port, where a first substantially straight radial edge of the first port and a second substantially straight radial edge of the second port substantially match corresponding substantially straight edges of the ducts **322** of the rotor **320**. In some embodiments, a second end cover **310** forms a third port and a fourth port, where a third substantially straight radial edge of the third port and a fourth substantially straight radial edge of the fourth port substantially match the corresponding substantially straight edges of the ducts **322** of the rotor **320**.
- (88) In some embodiments, at least one of the radial sidewalls **314** (e.g., leading radial sidewall) that form the HPIN port is configured to open relative to a duct **322** formed by the rotor **320** at least two degrees (e.g., four degrees, six degrees, eight degrees, ten degrees, etc.) later than at least one of the second radial sidewalls (e.g., leading radial sidewall) that form the HPOUT port opens relative to the duct **322**.
- (89) In some embodiments, an end cover **310** (e.g., that forms HPIN port) forms a fillet (or filet) (e.g., concave strip of material roughly triangular in cross section that rounds off an interior angle between two surfaces) between at least one of the radial sidewalls **314** and an inner sidewall **316**. In some embodiments, the end cover **310** that forms the HPOUT port may form the HPOUT port

without a fillet between the radial sidewalls **314** and the inner sidewall **316**. The fillet of the end cover **310** (e.g., that forms HPIN port) is configured to close the HPIN port relative to a duct **322** of the rotor **320** prior to the HPOUT port of the end cover that forms HPOUT port closes relative to the duct **322**.

(90) The fillet may be located at HPIN port exit inner diameter corner. The rotor **320** may form multiple circular rows of ducts **322** with outer rows being approximately trapezoidal in shape and the inner most row being triangular in shape. Due to the reduction in roundness from outer ducts to inner ducts, the inner most row of triangular shaped ducts may contribute more to mixing at HPOUT (per unit flow area) than ducts in the outer rows. By increasing the fillet radius at the inner diameter exit corner of the HPOUT port, mixing contribution from inner ducts may be reduced. A fillet at HPIN port bottom corner results in reduction of fluid (e.g., brine concentration, the fluid with which pressure is being exchanged) exiting out of the HPOUT row from the inner row of ducts **322**.

(91) In some embodiments, the radial sidewalls **314** of the end cover **310** (e.g., that forms LPIN port and HPOUT port) each form a ramp (e.g., sloped sidewall, sidewall that is not perpendicular to the face of the end cover) (e.g., see FIGS. 3A-E) and the radial sidewalls **314** of the other end cover forms HPIN port (e.g., and LPOUT port) without forming ramps.

(92) In some embodiments, the radial sidewalls **314** of the end cover **310** (e.g., that forms LPIN port and HPOUT port) each form a ramp (e.g., sloped sidewall, sidewall that is not perpendicular to the face of the end cover) (e.g., see FIGS. 3A-E) and the radial sidewalls **314** of the other end cover forms HPIN port (e.g., and LPOUT port) each form a ramp.

(93) The first radial sidewalls include a leading sidewall and a trailing sidewall on the HPIN port that each have a corresponding ramp angle varying from about 30 degrees to about 70 degrees measured with respect to a face of the rotor **320**.

(94) In some embodiments, leading radial sidewall and trailing radial sidewall on the LPIN port form a ramp in direction of rotation. The ramp has a ramp angle varying from about 30 degrees to about 70 degrees measured with respect to a face of the rotor **320**.

(95) In some embodiments, the radial sidewalls **314** of the end cover **310** (e.g., that forms LPIN port and HPOUT port) form non-planar three-dimensional ramps (e.g., see ramps **317A-B** of FIG. 3E). In some embodiments, the first radial sidewalls form non-planar three-dimensional ramps defined by at least two helix at an innermost radius and outermost radius of the port. In some embodiments, a corresponding ramp of the non-planar three-dimensional ramps is defined by a spiral that has a pitch that is proportional to a radius at which the corresponding ramp is located to reduce incidence (e.g., to reduce incidence and to cause absolute velocity (c) to be similar at any radius of the port).

(96) The ramps **317** may be associated with velocity triangle **319**. Velocity triangle **319** may include the following: $c_{sub.ID} = (u_{sub.ID}) * \tan(\alpha_{sub.ID})$ $c_{sub.OD} = (u_{sub.OD}) * \tan(\alpha_{sub.OD})$ $\alpha_{sub.ID}$ = kidney ramp angle at the inside diameter (ID) of the kidney (port **312** of end cover **310**) $\alpha_{sub.OD}$ = kidney ramp angle at the outside diameter (OD) of the kidney (port **312** of end cover **310**) $u_{sub.ID}$ = tangential velocity at inner diameter of port $u_{sub.OD}$ = tangential velocity at outer diameter of port $c_{sub.ID}$ = absolute velocity at inner diameter of port $c_{sub.OD}$ = absolute velocity at outer diameter of port

(97) Flow exiting the rotor **320** of a PX **300** may have a combination of axial and tangential components of flow velocity. The axial component is a function of flow rate and the tangential component is a function of rotor speed and radial position of the duct **322**. In some embodiments, the outlet kidneys (e.g., port **312** of end cover **310**) are shaped to allow for smooth transition of this three-dimensional (3D) velocity field from the rotor **320** to a one-dimensional (1D) velocity field along the stator conduits (e.g., ports **312** of end cover **310**). Otherwise, shock losses may result due to flow separation and eddy formation in the outlet kidneys (e.g., ports **312** of end cover **310**). The exit ramps can have a single ramp angle based on the design flow rate, target RPM, a median radial

position of the duct **322**, and/or the like. The outlet kidneys (e.g., ports **312** of end cover **310**) can also have dual ramps at the beginning and end of the outlet kidneys with similar considerations. In some embodiments, a 3D-shaped outlet kidney (e.g., port **312** of end cover **310**) may have a ramp angle that varies continually with the radial position. To overcome manufacturing difficulties of the 3D ramps, an injection molded plastic insert with a particular shape can be embedded into the outlet kidneys (e.g., port **312** of end cover **310**).

(98) In some embodiments, the end cover **310** includes an insert **330** disposed in the HPIN port, where the insert is configured to guide flow and to reduce flow incidence at the rotor **320** (e.g., see FIG. 3F).

(99) In some embodiments, the end cover **310** forms a first spot face **340** proximate the HPIN port (e.g., without forming a spot face at the HPIN port to reduce mixing of corresponding fluid between the HPIN port and the HPOUT port) and forms a second spot face **340** proximate the LPOUT port (e.g., see FIGS. 3L-O). In some embodiments, the first end cover **310** forms a second spot face proximate the LPOUT port without forming a spot face at the LPIN port to reduce mixing of corresponding fluid between the LPIN port and the LPOUT port.

(100) In some embodiments, the pressure exchanger **300** (e.g., PX **40**) is configured to use the second fluid provided via the LPIN port and/or HPOUT port as bearing fluid and/or centerbore fluid (e.g., see FIGS. 3I-K).

(101) FIG. 3P illustrates PX **300**, according to certain embodiments. Rotor **320** may include inserts **330** disposed in ducts **322**. The inserts may provide a length to diameter ratio of about 5 to about 10. The inserts may be honeycomb-shaped inserts. The inserts may be honeycomb-shaped flow straighteners that are press-fit or shrunk-fit into the ducts. The inserts **330** may maximize the length to diameter (L/D) ratio and the roundness of ducts **322**. The inserts **330** may be honeycomb flow straighteners in the ducts **322** to reduce mixing of fluids.

(102) Turbulence of unsteady fluid flow in the ducts **322** may cause (e.g., may be a major source of) intra-duct mixing. This results in skewing & stretching of the “mixing zone” between the two fluids in the duct **322**. In some embodiments, to mitigate this, the aspect ratio of duct geometry (e.g., length to diameter—L/D) is to be increased by inserting inserts **330** (e.g., honeycomb flow straighteners) into the ducts **322**. The flow straighteners can be press-fit or shrunk-fit into the ducts **322** or attached to the duct walls mechanically or adhesively (e.g., gluing). The insert **330** (e.g., flow straightener) can have shapes other than honeycomb and the insert **330** may divide a duct of a certain L/D ratio into multiple ducts of larger L/D ratio. Breaking a duct into multiple small ducts also reduces cavitation and noise by temporally spreading out pressurization and depressurization events at kidney openings (e.g., openings of ports **312** of end covers **310**).

(103) In some embodiments, pressure exchanger **300** is used to minimize mixing of two fluids inside the pressure exchanger **300**. In the pressure exchanger **300**, liquid-to-liquid pressure exchange may take place via an oscillating “liquid plug” in the rotor ducts. The “liquid plug” may not be impenetrable and conventionally a small amount of mixing may occur between the two fluids exchanging their pressure energy. Mixing may be dependent on one or more factors, such as travel distance of the “liquid plug” (e.g., portion of the duct **322** of the rotor **320** traverses), turbulence, diffusion, jetting, entry and/or exit losses of the rotor **320**, etc. Efficiency of the pressure exchanger **300** may be directly proportional to the travel distance of the “liquid plug.” Mixing of a first fluid and a second fluid in the pressure exchanger **300** may be inversely proportional to the travel distance (and to the efficiency in the pressure exchanger **300**). In some examples, less travel distance results in less mixing and lower efficiency. In some examples, more travel distance results in more mixing and higher efficiency (e.g., features that attempt to increase efficiency by increasing travel distance conventionally also increase mixing within the pressure exchanger). In some applications (e.g., a SWRO plant), efficiency of a pressure exchanger **300** is to be maximized and mixing within the pressure exchanger **300** is to be minimized to reduce specific energy consumption (e.g., of the SWRO plant).

(104) Pressure exchanger **300** may include one or more features configured to reduce mixing within the pressure exchanger **300** without having a negative impact on efficiency of the pressure exchanger **300** (e.g., minimizing decrease in efficiency, maintaining efficiency, increasing efficiency compared to pressure exchangers that do not have those features).

(105) The features of pressure exchanger **300** (e.g., see end covers **310** of FIGS. 3A-H) may provide reduced rotor incidence losses (e.g., two ramps at entrance of port **312** to reduce incidence). Incidence may be an angle between ideal c velocity vector and actual c velocity vector.

(106) Flow within ducts **322** of rotor **320** has both an axial velocity component (e.g., $c(m)=w$ velocity component with two ramps of FIG. 3C) and a tangential velocity component (e.g., u -velocity component on FIG. 3C). The axial component may remain constant at any axial plane of the rotor **320**, whereas the tangential component varies linearly with the radial coordinate. As the flow enters the rotor **320** through HPIN port and LPIN port (e.g., HPIN and LPIN kidneys), the flow may be accelerated to a velocity with a combination of axial and tangential velocity components, at all radii just before the fluid enters the ducts **322** of the rotor **320**. Conventionally, shock losses due to oblique incidence can occur, resulting in flow separation and mixing due to enhanced eddy formation. End cover **310** forms ports **312** configured to guide the flow for a given nominal flow rate and a target RPM of rotor **320** that results in substantial reduction in mixing and reduction in pressure loss due to sudden change in flow direction.

(107) In some embodiments, end cover **310** includes two ramps (e.g., a ramp at each radial sidewall **314**) to provide substantial tangential flow velocity component in addition to the axial flow velocity component. In some embodiments, the end cover **310** has 3D ramps (e.g., further kidney geometry optimization) to provide smooth entry of fluid from port **312** to ducts **322** (e.g., see right side of FIG. 3C)

(108) In some embodiments, an insert **330** (e.g., made of one or more components) may be embedded into the port **312** (e.g., see FIG. 3F). The insert **330** may be made of Alumina ceramic, plastics such as polycarbonate, fiber reinforced polytetrafluoroethylene (PTFE) and/or polyetheretherketone (PEEK), and/or the like.

(109) End cover **310** of FIG. 3H may reduce effective width of the port **312** (e.g., kidney) by opening HPIN port **312** a few degrees later than baseline and thus reduces efficiency.

(110) HPIN port may close earlier than HPOUT port (e.g., radial sidewalls **314** of HPIN port are closer to each other than radial sidewalls **314** of HPOUT). This may reduce mixing at HPOUT port caused by fluid inertia. Mixing at HPOUT port is undesirable when a pressure exchanger is used for energy recovery in particular applications (e.g., in an SWRO plant). The flow in the duct **322** of the rotor **320** accelerates as the duct **322** traverses across the HP ports. Average duct flow velocity starts from near-zero when the duct **322** opens to the HP ports and reaches a maximum just as it begins to exit the ports (e.g., kidneys).

(111) Conventionally HPIN port and HPOUT port close at the same time (e.g., are the same size, radial sidewalls are same distance from each other). This may result in causing the duct flow to come to an abrupt halt during the short time for the fluid in the duct **322** to exit the port **312**. Since the duct flow velocity is changed rapidly from a max value to near-zero in a very short time, inertia of the duct fluid may cause a large spike in local duct pressure. This may result in a fluid jet shooting out into the HPOUT port through the rapidly closing opening between the duct **322** and the HPOUT port. At that moment, the mixing zone (e.g., “liquid plug”) may have traveled the maximum extent through the duct and may be proximate (e.g., very close) to the HPOUT port. Jets into HPOUT port may transport fluid from the mixing zone as well as some HPIN fluid behind the mixing zone into the HPOUT port, increasing undesirable mixing at HPOUT port.

(112) The present disclosure may mitigate this mixing increase by closing the HPIN port a few degrees prior to the HPOUT port. This results in peak velocities being reached before the duct **322** closes to HPOUT port. Duct flow is decelerating when the fluid in the duct **322** exits the HPOUT port causing the pressure spike due to fluid inertia to decrease substantially. Jetting of mixing zone

fluid into HPOUT port is greatly reduced, resulting in reduced mixing at HPOUT port.

(113) HPIN port may open later than HPOUT port. Delaying the HPIN port opening after the HPOUT port biases the mean location of the two-fluid interface closer to the HPIN port than the LPIN port. This helps move the interface away from HPOUT and reduces mixing of HPIN at HPOUT. This feature can be selectively utilized when reducing mixing is a higher priority and slight tradeoff in efficiency is acceptable. Delaying HPIN opening a few degrees with respect to HPOUT can move the interface away from HPOUT and this may reduce mixing at HPOUT.

(114) The kidney exit ramps (e.g., radial sidewalls **314**) of the end covers **310** may be configured to guide flow into HPOUT and LPOUT ports while minimizing shock losses at exit (e.g., shock losses due to flow separation).

(115) Flow exiting ducts **322** of rotor **320** into outlet ports (e.g., outlet ports, HPOUT port and LPOUT port) has both axial and tangential components of velocity. Axial velocity component may not vary much with radius, but tangential velocity component scales linearly with radius. Outlet port walls may be configured to provide duct flow smoothly exiting through the ports without any separation. The sidewalls of the end cover **310** that form the ports **312** may be configured to accommodate the varying ratio of tangential and axial components of velocity with radius. By preventing flow separation at exit, both differential pressure (DP) losses and mixing losses are reduced (e.g., mixing zone may be closest to the duct exit at outlet ports).

(116) Spot faces **340** may be located to reduce mixing (e.g., see FIGS. 3L-O). Spot faces **340** may be used on ports **312** to gradually pressurize or depressurize a duct **322** of a rotor **320** (e.g., instead of abrupt pressure equalization through high velocity jetting into or out of the rotor duct). Spot faces **340** reduce the jetting velocity due to pressure differential between the duct **322** and the port **312** by adding resistance to flow through the narrow gap. In some embodiments, the spot faces **340** are located on HPOUT port without being located in HPIN port. This may compensate for the depth of the spot face **340** and pressure equalization may be achieved by jets of HPOUT fluid entering the low-pressure rotor duct instead of HPIN fluid. The “liquid plug” may be farthest away from HPOUT port when the pressure equalization occurs and mixing is reduced.

(117) In some embodiments, second fluid (e.g., exiting HPOUT port, HPOUT fluid) may be used as bearing fluid and/or centerbore fluid. In a pressure exchanger **300** (e.g., rotary pressure exchanger), first fluid (e.g., HPIN fluid) or second fluid (e.g., HPOUT fluid) may be used as bearing fluid for radial and axial bearings. If the pressure differential between HPIN and HPOUT fluids meets a threshold amount (e.g., is small), bearing performance (e.g., stiffness and load capacity) may not change appreciably with choice of either of the fluids. If the pressure differential between HPIN and HPOUT fluids meets a threshold amount (e.g., is small), HPOUT fluid may be used as the bearing fluid and may reduce mixing at HPOUT port. By feeding the radial and axial bearings with HPOUT fluid, HPIN fluid may be isolated preventing the chance of increased mixing through leakage from the bearing gaps.

(118) HPOUT fluid may be used to feed pressure exchanger radial bearings and to fill the centerbore of the rotor (e.g., see FIGS. 3I-J).

(119) Referring to FIG. 3I, PX **300** may include a fluid bypass **350** (e.g., HPOUT fluid bypass), according to certain embodiments. Fluid may enter between the housing **352** and end cover **310** (e.g., LPIN end cover) and flow between housing **352** and sleeve **301** and via an opening in sleeve **301** (e.g., to feed pressure exchanger radial bearings and/or to fill the center bore). In some embodiments, fluid bypass **350** (e.g., HPOUT fluid bypass) may go around a gasket (e.g., O-ring) between housing **352** and end cover **310** to feed the PX radial bearings through a hole in the sleeve.

(120) Referring to FIG. 3J, PX **300** may include a fluid bypass **350** (e.g., HPOUT fluid bypass) and/or slot **354**, according to certain embodiments. The fluid bypass **350** may be an HPOUT fluid bypass of gasket (e.g., O-ring) and may feed the PX radial bearings through a hole in the sleeve. Slot **354** in end cover **310** (e.g., HPOUT end cover) may be used to communicate HPOUT fluid to center bore **356** (e.g., central portion of rotor **320**, between rotor **320** and shaft).

(121) In some embodiments, the length to diameter (L/D) ratio and the roundness of the ducts **322** are maximized. Mixing in a rotor duct may be a strong function of turbulence, the influence of which can be reduced significantly by controlling the L/D ratio and the roundness of the ducts. Inserts may be added to the rotor **320** to achieve a desired duct shape, achieve near “liquid plug” flow, and reduce mixing in the rotor ducts.

(122) The present disclosure may be employed in rotary pressure exchangers to reduce mixing of the two fluids across which pressure energy is being exchanged.

(123) The present disclosure may include features such as ramps (e.g., kidney ramps), spot faces **340**, filets, etc. (e.g., to reduce mixing).

(124) The present disclosure reduces the mixing of fluids in a pressure exchanger and improves overall energy consumption per unit volume (e.g., of potable water produced in a SWRO plant). The present disclosure is applicable for different pressure exchanger (e.g., isobaric pressure exchanger) applications, such as SWRO, sCO₂, industrial wastewater, etc. The present disclosure may also be applicable for pressure exchanger architectures, such as a PX with rotor-sleeve or sleeveless pressure exchanger (e.g., rotor with center-post) and motorized or non-motorized PX. The present disclosure can be applied appropriately for different applications to reduce mixing of LPIN into LPOUT as well.

(125) FIGS. **4A-K** illustrate components of pressure exchangers **300** (e.g., PXs **40** of FIGS. **2A-E**, pressure exchanger **300** of one or more of FIGS. **3A-P**), according to certain embodiments. Features in FIGS. **4A-K** that have similar names and/or reference numbers as features in one or more of FIGS. **1A-2E** and/or one or more of FIGS. **3A-P** may include the same or similar structure, material, functionality, and/or the like as features in one or more of FIGS. **1A-2E** and/or one or more of FIGS. **3A-P**. In some embodiments, one or more features of PX **300** of FIGS. **4A-K** increase efficiency of PX **300**.

(126) In some embodiments, a pressure exchanger **300** includes a rotor **320** and end covers **310**. The rotor **320** is configured to rotate to exchange pressure between a first fluid at a first pressure and a second fluid at a second pressure. The rotor **320** forms ducts **322** that are routed from a first distal end of the rotor **320** to a second distal end of the rotor **320**.

(127) In some embodiments, radial edge of kidney entrance and exits of end cover **310** may matching that of the rotor **320**.

(128) In some embodiments, PX **300** uses a streamlined (lofted) spacer in end cover **310**.

(129) In some embodiments, non-metallic PVC spacer is used in PX **300** to handle axial thrust and to avoid fretting corrosion between ceramic end cover **310** and metallic interconnect.

(130) In some embodiments, radial and axial bearing clearances are optimized for target circumference groove pressure.

(131) In some embodiments, the ports of end cover **310** are maximized to be substantially the same size as one duct **322** plus one wall sealing area of rotor **320** to reduce difference in pressure (DP) losses.

(132) In some embodiments, pressure loads are substantially balanced on both rotor faces of rotor **320**.

(133) In some embodiments, duct shape of ducts **322** substantially matches ports (e.g., kidneys) of end covers **310** (e.g., trapezoidal or triangular instead of circular).

(134) In some embodiments, an end cover **310** (e.g., a first end cover) is disposed at a distal end (e.g., the first distal end) of the rotor **320** and the end cover **310** forms a HPIN port configured to provide the first fluid at the first pressure into the ducts **322** and forms a LPOUT port configured to receive the first fluid from the ducts **322** at a third pressure that is lower than the first pressure. One or more sidewalls (e.g., radial sidewalls **314**) of the end cover that form the HPIN port and/or the LPOUT port are substantially planar (e.g., straight edge **401**, not curved edge **402**) (e.g., see FIG. **4B**). In some embodiments, the one or more sidewalls of the first end cover **310** that form the HPIN port have a substantially straight edge **401** that substantially matches a corresponding

substantially straight edge of a duct **322** formed by the rotor **320**.

(135) In some embodiments, an end cover **310** is disposed at a distal end (e.g., second distal end) of the rotor **320** and the end cover forms a LPIN port configured to provide the second fluid at the second pressure into the ducts **322** and forms a HPOUT port configured to receive the second fluid from the ducts **322** at a fourth pressure that is higher than the second pressure.

(136) In some embodiments, the end cover **310** (e.g., that forms HPIN port and LPOUT port, that forms LPIN port and HPOUT port) includes substantially planar radial sidewalls **314** (e.g., non-curved radial sidewalls **314**), an inner sidewall **316**, and an outer sidewall **318** that form the port **312** (e.g., HPIN port, HPOUT port). The first substantially radial sidewalls **314** may be disposed between a center of the end cover **310** and a perimeter of the end cover **310**. The inner sidewall **316** may be proximate the center of the end cover **310** and the outer sidewall **318** may be proximate the perimeter of the end cover **310**.

(137) The rotor **320** may form ducts **322** in concentric rows (e.g., see FIGS. 4C-D). A ratio of a number of the concentric rows to inches of diameter of the rotor **320** may be about 0.3 to about 0.45 (e.g., from about 0.375 to about 0.42). The rotor **320** may form the ducts **322** in at least three concentric rows.

(138) The pressure exchanger **300** may include a first interconnect **420** configured to provide the first fluid to the HPIN port of the end cover **310**, a first spacer **410** (e.g., see FIG. 4E) disposed in the end cover **310** between the first interconnect and the HPIN port (e.g., see FIG. 4F). The pressure exchanger **300** may include a second interconnect **420** configured to receive the first fluid from the LPOUT port of the end cover **310** and a second spacer **410** disposed in the first end cover between the LPOUT port and the second interconnect (e.g., see FIG. 4F). In some embodiments, the first spacer and/or the second spacer are a thermoplastic material. In some embodiments, the first spacer and/or the second spacer are polyvinyl chloride (PVC).

(139) FIGS. 4G-4H illustrate PXs **300**, according to certain embodiments. In some embodiments, a PX **300** may have bearing stiffness tuned to center the rotor **320** and to reduce leakage by optimizing diametral clearance **303** (e.g., radial clearance) and axial clearance **304** (e.g., axial bearing clearance, a height difference between the sleeve and the rotor or between the center post and the rotor) for target circumferential groove pressure.

(140) In PX **300** (e.g., a rotary pressure exchanger), the rotor **320** is separated from the stator (e.g., end cover **310**, sleeve **301**) with a small clearance both radially (e.g., diametral clearance **303** between the sleeve **301** and rotor **320** or center post **302** and rotor **320**) and axially (e.g., axial clearance **304** between end covers **310** and rotor **320**). During operation, the rotor **320** is suspended in the clearance by the stiffness of the fluid film in the radial and axial bearings generated by hydrodynamic and hydrostatic effects respectively. For minimal bearing flows (leakage loss), the rotor **320** is to have minimal clearances and the rotor **320** is to be centered axially (e.g., with smallest axial eccentricity). Minimal clearances are set by manufacturing and material stiffness limitations. Minimal eccentricity can be achieved by maintaining the ratio between the diametral clearance **303** and axial clearance **304** within a narrow range of about 1 to about 3.5. Such an arrangement may provide an intermediate plenum pressure (e.g., the optimal intermediate plenum pressure) (e.g., in circumferential plenum **305**) between the two bearings resulting in lower axial rotor eccentricity (e.g., lowest axial rotor eccentricity) and in so doing, minimize leakage loss.

(141) In some embodiments, a ratio of diametral clearance **303** (e.g., diametric clearance, diametric clearance, radial clearance, etc.) to axial clearance **304** is about 1 to about 3.5. The diametral clearance **303** is between the rotor **320** and a sleeve **301** or between the rotor **320** and a center post **302**. The axial clearance **304** is between the rotor **320** and an end cover **310**.

(142) FIG. 4I illustrates a rotor **320**, according to certain embodiments. FIG. 4J illustrates an end cover **310** according to certain embodiments. FIG. 4K illustrates a PX **300** with rotor **320** superimposed on end cover **310**, according to certain embodiments. As shown in FIGS. 4I-K, sealing angle **313** between ports **312** of end cover **310** may be at least the same as the duct angle

323 plus duct wall angle **324** of rotor **320**.

(143) In some embodiments, angular spacing between the HPIN port and the LPOUT port of the first end cover **310** substantially matches corresponding angular spacing between a first leading radial sidewall of a first duct **322** of the rotor **320** and a second leading radial sidewall of a second duct **322** that is adjacent to the first duct **322**.

(144) Fluid flow in the ducts of a rotary pressure exchanger may be unsteady with rapid acceleration and deceleration in both directions during each rotation. This results in a pressure loss across the rotor required to overcome fluid inertia. Apart from reducing rotor speed (which can result in increased mixing), this inertial pressure loss can also be minimized by maximizing the flow area of the kidneys (e.g., ports **312** of end cover **310**). This results in lower acceleration for the same flow rate which provides a reduction in inertial pressure loss. To prevent direct communication from one kidney to the next (e.g., between ports **312**), a minimum amount of sealing angle of end cover **310** may be maintained substantially equivalent (e.g., equivalent) to the angular spacing between the ducts **322** (duct angle **323** plus duct wall angle **324**).

(145) The present disclosure may optimize a pressure exchanger **300** (e.g., isobaric pressure exchanger) for efficiency, mixing, cavitation, sound, and cost.

(146) Conventionally, features that attempt to increase efficiency by increasing travel distance end up increasing mixing inside the pressure exchanger. The present disclosure may maximize efficiency at a similar travel distance as conventional pressure exchangers without having a negative impact on mixing performance (e.g., by maintaining the same amount of mixing, by decreasing mixing, etc.).

(147) The present disclosure may be used in different pressure exchanger (e.g., isobaric pressure exchanger) applications (e.g., using a pressure exchanger for energy recovery), such as SWRO, sCO₂, industrial waste water, etc. The present disclosure may be used in different pressure exchanger architectures, such as a pressure exchanger with rotor-sleeve or sleeveless pressure exchanger (e.g., rotor with center-post) and motorized or non-motorized pressure exchanger.

(148) The present disclosure may be used to reduce fluid inertial pressure loss using radial edge of the port **312** (e.g., kidney) entrance and exits. Conventionally, a substantially amount of the pressure loss in the rotor **320** occurs due to sudden acceleration and deceleration of flow inside the rotor **320**. The differential pressure to accelerate the flow inside the duct **322** of the rotor **320** (e.g., pressure lost in accelerating the flow inside the rotor ducts in terms of dQ/dt) may be given as shown in Equation 1.

$\Delta P_{\text{sub.fluid_inertia}} = (\rho_{\text{sub.fluid}} * L_{\text{sub.duct}} / A_{\text{sub.duct}}) * (dQ_{\text{sub.duct}} / dt)$ Equation 1:

$Q_{\text{sub.duct}}$ is the flow through each duct **322**. $dQ_{\text{sub.duct}} / dt$ is the rate of change of flow in the duct **322** (e.g., acceleration, deceleration). $L_{\text{sub.duct}}$ is the length of the duct **322**. $A_{\text{sub.duct}}$ is the area of the duct **322**. dt is the time the duct **322** is open to the port **312** (e.g., kidney).

$dP_{\text{sub.fluid_inertia}}$ (e.g., $\Delta P_{\text{sub.fluid_inertia}}$) is the pressure loss in the fluid due to acceleration and/or deceleration.

(149) Since $dt = \omega * d\theta$, Equation 1 can be re-written as Equation 2 (e.g., pressure lost in accelerating the flow inside the rotor ducts in terms of $dQ/d\theta$).

$\Delta P_{\text{sub.fluid_inertia}} = (\rho_{\text{sub.fluid}} * L_{\text{sub.duct}} / A_{\text{sub.duct}} / \omega) * (dQ_{\text{sub.duct}} / d\theta)$ Equation 2:

(150) Per Equation 2, the $dP_{\text{sub.fluid_inertia}}$ (e.g., $\Delta P_{\text{sub.fluid_inertia}}$) can be decreased by reducing the $dQ/d\theta$ term.

(151) Conventional end covers may have curved-edge sidewalls that form the ports. In some embodiments, end covers **310** of the present disclosure include substantially planar radial sidewalls **314** (e.g., radial-edge end covers **310**). Using a radial-edge port may cause an effective kidney angle for the middle and outer-most ducts to be increased as compared to curved-edge ports (e.g., by about 26% for outer-most duct). The radial edge of the port **312** may cause the trapezoidal shape of the port **312** to match the trapezoidal shape of ducts **322** of rotor **320**.

(152) In some embodiments, fluid inertial pressure loss may be reduced by maximizing rotor flow

with selection of number and shape of rotor ducts. In a rotor **320**, duct passage-ways help convey flow from the inlet to the outlet ports. Most amount of pressure losses may occur in the rotor **320**. Increasing the rotor duct flow area may reduce pressure loss, without having a negative impact on mixing. The constraint may be that stress are to be below the material strength of the rotor **320**.

(153) In some embodiments, the rotor **320** has trapezoidal shaped ducts **322** with large fillets to make use of available area by maximizing duct flow area and preventing high stress concentrations (e.g., see FIG. **4D**).

(154) In some embodiments, the number of concentric rows is selected to maximize the total duct flow area, while also minimizing the maximum duct area and meeting the strength criteria of the material. This also helps achieve effective hydraulic diameter similar to an equivalent circle (e.g., prevents high aspect ratio ducts).

(155) In some embodiments, the rotor **320** has an odd number of ducts **322** (e.g., a non-even number of ducts). This prevents opening of a duct **322** to symmetrically opposite ports **312** at the same time which reduces noise and vibration of the pressure exchanger **300**.

(156) In some embodiments, the rotor **320** has staggered ducts which reduces the total amount of duct volume that pressurizes and/or depressurizes at a time which reduces noise and vibration of the pressure exchanger.

(157) In some embodiments, the rotor **320** has at least two concentric rows of ducts **322**. In some embodiments, the rotor **320** has at least three concentric rows of ducts **322**.

(158) In some embodiments, a spacer (e.g., lofted spacer, streamlined spacer) is used to provide the flow into the HPOUT port and the LPOUT port is guided to minimize shock losses at exit due to flow separation.

(159) Flow exiting rotor ducts into outlet ports has both axial and tangential components of velocity. Axial velocity component may not vary much with radius of the rotor **320**. Tangential component scales linearly with radius. Outlet port (e.g., LPOUT port, HPOUT port) walls may be designed so that the duct flow exits smoothly through the ports **312** without separation. The port sidewalls (e.g., kidney walls) may be configured to accommodate the varying ratio of tangential and axial components of velocity with radius. By preventing flow separation at exit, both pressure differential losses and mixing losses are reduced. The mixing zone may be closest to the duct exit at outlet ports (e.g., outlet kidneys) of the end covers **310**.

(160) A spacer **410** (e.g., streamlined spacer, streamlined lofted spacer) may be used to reduce pressure loss. In some embodiments, a spacer may be used to eliminate the use of a thrust ring and/or to reduce pressure losses due to sudden change in area between LP ports and interconnects. A spacer **419** (e.g., lofted spacer) may be used to streamline flow and to avoid pressure loss due to sudden change in area. In some embodiments, PX **300** may include a spacer **410** that has a lofted shape that transitions from a round shape of an interconnect to a non-round shape of a corresponding port of the first end cover **310**.

(161) A non-metallic spacer **410** (e.g., non-metallic streamlined spacer) may be used to handle axial thrust and to avoid fretting corrosion between ceramic end cover **310** and metallic interconnects (e.g., end covers **310** used with interconnects without a spacer **410** may have fretting corrosion issues). A spacer **410** (e.g., streamlined spacer) may be made out of polyvinyl chloride (PVC) (e.g., to eliminate need for a thrust ring used in conventional pressure exchangers). The thrust ring transmits axial force from the ceramic cartridge to the housing through the seal plate and bearing plate. The use of a spacer **419** made of PVC (e.g., injected molded to reduce cost) acts as buffer material between metallic LP interconnects and the ceramic end cover **310**. This allows the freedom to select the LP interconnect material to any material (e.g., 2507 super duplex, AL6XN, etc.) compatible with the fluid of the application (e.g., seawater application) and for the rated pressure.

(162) In some embodiments, leading radial sidewalls and trailing radial sidewalls of the LPOUT port and the HPOUT port are not axial. The leading radial sidewalls and the trailing radial sidewalls

of the LPOUT port and the HPOUT port may be inclined at an angle that is substantially proportional to rotor revolutions per minute. The angle may be about 30 degrees to about 70 degrees.

(163) Fluid acceleration and deceleration (e.g., fluid inertia) losses may be reduced by: increasing time available to fill rotor ducts by increasing effective angular extent of the port by making port edges radial (e.g., substantially planar radial sidewalls **314**); minimizing peak fluid velocity (e.g., V_{max}) inside the rotor **320** by maximizing rotor flow area while meeting stress and manufacturing constraints.

(164) In some embodiments, a spacer **410** (e.g., lofted spacer) helps reduce fretting corrosion between super duplex or stainless steel metals and Aluminum oxide (Al.sub.2O.sub.3) and may streamline flow in and out of the ports which reduces frictional fluid losses. This may avoid using more exotic materials for the interconnect **420** such as titanium. The spacer (e.g., lofted spacer) may combine three functions of streamlining flow and reducing fluid loss due to sudden area change, acts as a compliant material between metal interconnect **420** and brittle ceramic, and also handles the net thrust of the cartridge due to pressure imbalance. This may prevent use of a separate thrust ring.

(165) Features, such as trapezoidal ducts **322**, odd number of ducts **322**, at least three concentric rows of ducts **322** while maintaining single duct volume at a minimum, and/or the like may be used to maximize rotor efficiency.

(166) FIGS. 5A-J illustrate components of pressure exchangers **300** (e.g., PXs **40** of FIGS. 2A-E, pressure exchanger **300** of one or more of FIGS. 3A-4K), according to certain embodiments. Features in FIGS. 5A-J that have similar names and/or reference numbers as features in one or more of FIGS. 1A-2E and/or one or more of FIGS. 3A-4K may include the same or similar structure, material, functionality, and/or the like as features in one or more of FIGS. 1A-2E and/or one or more of FIGS. 3A-4K. In some embodiments, one or more features of PX **300** of FIGS. 5A-J reduce cavitation and/or noise of PX **300**.

(167) A pressure exchanger includes a rotor **320** and end covers **310**. The rotor **320** is configured to rotate to exchange pressure between a first fluid at a first pressure and a second fluid at a second pressure. The rotor **320** forms ducts **322** that are routed from a first distal end of the rotor **320** to a second distal end of the rotor **320**.

(168) An end cover **310** (e.g., disposed at a distal end of the rotor **320**) forms an HPIN port configured to provide the first fluid at the first pressure into the ducts **322** and forms an LPOUT port configured to receive the first fluid from the ducts **322** at a third pressure that is lower than the first pressure.

(169) In some embodiments, PX **300** includes a spot face at LPIN open instead of LPOUT open (e.g., to cause depressurization of HP duct to occur at LPIN which is at a higher pressure than LPOUT).

(170) In some embodiments, duct opening to the spot face at duct ID instead of duct OD (e.g., adjusting angle of spot face edge or curving the spot face wall towards or away from duct).

(171) In some embodiments, spot faces with radial extent shorter than that of the duct **322** (e.g., reduces the diameter at which duct will open to kidney, reduces tangential velocity and cavitation potential).

(172) In some embodiments, PX **300** includes split spot faces (e.g., at one or more end covers **310**). There may be one split spot face per row of ducts or combined for a set of ducts. This may tune spot face for each row of ducts **322** separately.

(173) In some embodiments, PX **300** has optimized duct volume and staggering ducts in adjacent rows.

(174) In some embodiments, spot face geometries may have: (a) substantially constant depth spot face; (b) substantially linear depth spot face; (c) step out EC (end cover); and/or (d) pre-pressurization hole.

(175) In some embodiments, spot face angular extent, radial extent, and ramp angle may be optimized for different fluids and operating conditions.

(176) In some embodiments, rotor **320** may include chamfers on trailing edge of rotor duct walls to increase coefficient of discharge (Cd).

(177) An end cover **310** (e.g., disposed at a distal end of the rotor **320**) forms an LPIN port configured to provide the second fluid at the second pressure into the ducts **322** and forms an HPOUT port configured to receive the second fluid from the ducts **322** at a fourth pressure that is higher than the second pressure (e.g., FIG. 5B). The end cover **310** forms a first spot face **340** proximate the LPIN port and a second spot face **340** proximate the HPOUT port.

(178) In some embodiments, an end cover **310** forms the LPOUT port without forming a spot face **340** proximate the LPOUT port and forms the HPIN port without forming a spot face **340** proximate the HPIN port (e.g., FIG. 5A).

(179) In some embodiments, for SWRO, a spot face **340** is formed proximate LPIN port and a spot face **340** is formed proximate HPOUT port to minimize mixing at HPOUT port (e.g., without forming a spot face at LPOUT which could cause cavitation since LPOUT pressure is below a threshold pressure).

(180) In some embodiments, for refrigeration, a spot face **340** is formed proximate LPOUT port and a spot face **340** is formed proximate HPOUT port to reduce mixing at LPOUT port and HPOUT port (e.g., LPOUT pressure is above a threshold pressure so risk of cavitation is low).

(181) In some embodiments, end cover **310** forms a chamfer **311** at the LPIN port that is configured to change a corresponding angle of the first spot face **340** for each concentric row of the ducts of the rotor **320** (e.g., see FIG. 5C). The chamfer **311** at the LPIN port is configured to change a corresponding angle for each duct to be exposed to the spot face to change an amount of pressurization (e.g., pressurization and/or depressurization) of different rows of ducts.

(182) In some embodiments, the first spot face **340** has a radial extent that is shorter than duct radial extent of a corresponding duct **322** formed by the rotor **320** (e.g., see FIGS. 5D-F).

(183) In some embodiments, the first spot face **340** (e.g., split spot face) includes a first recess associated with a first concentric row of the ducts **322** and a second recess associated with a second concentric row of the ducts **322** (e.g., see FIG. 5G).

(184) In some embodiments, the rotor **320** forms the ducts **322** in concentric rows that are staggered (e.g., see FIGS. 5H-I).

(185) In some embodiments, duct opening of the rotor **320** to a corresponding end cover spot face is at an inside duct diameter by adjusting angle of spot face edge or curving spot face wall respective to the duct opening.

(186) In some embodiments, the end cover **310** (e.g., that forms LPIN port and HPOUT port) forms a pre-pressurization hole proximate the first spot face **340**. The pre-pressurization hole is configured to one or more of pressurize or depressurize at least a portion of the pressure exchanger proximate the end cover **310** (e.g., that forms LPIN port and HPOUT port).

(187) FIG. 5J illustrates a rotor, according to certain embodiments. In some embodiments, the rotor **320** forms chamfers **500** on trailing edge rotor duct walls. Chamfers **500** on trailing edge of rotor duct walls may increase coefficient of discharge (Cd) of rotor **320**.

(188) Adding a chamfer or fillet on the trailing edge of a rotor duct wall may increase the effective area (coefficient of discharge) for depressurization jets from the high pressure duct into the low pressure kidney. This increase in effective area reduces the jet velocity and the consequent drop in local fluid static pressure, which may reduce cavitation potential. Similarly, a chamfer or fillet may be employed on the leading edge of a rotor duct wall to increase the effective area (coefficient of discharge) for pressurization jets from the kidney into the duct **322**. In some embodiments, these features, by reducing the peak jet velocity, also reduce noise and vibration.

(189) The present disclosure may provide cavitation, noise, and vibration control in a pressure exchanger **300** (e.g., rotary isobaric pressure exchanger).

(190) In a pressure exchanger (e.g., rotary isobaric pressure exchanger), rotating ducts carry high pressure fluids from high pressure kidneys (ports) towards low pressure kidneys and also low pressure fluids from low pressure kidneys to high pressure kidneys. The duct fluids undergo rapid pressurization or depressurization whenever the duct fluids approach a set of kidneys. The frequency and the rate of pressurization and/or depressurization depends on the rotor RPM, the number of pressure exchange cycles per revolution and the pressure differential between HP and LP kidneys. This rapid pressurization and/or depressurization results in high velocity fluid jets generating noise and producing flow and pressure pulsations which increase vibration levels. This can produce vapor bubbles if the local fluid pressure (e.g., due to high velocity of the local fluid pressure) falls below the vapor pressure at that temperature. These vapor bubbles collapse as they travel to higher pressure regions resulting in void formation causing surrounding fluid to rush in producing very high localized pressure spikes. If these occur next to a solid wall, they can produce pitting damage, which accumulates over time. This can cause cavitation which also amplifies the noise and vibration levels.

(191) Spot faces **340** (e.g., channels, notches, recesses) may be used to control the rate of pressurization or depressurization of the rotor ducts **322** and thereby reduce noise, vibration, and risk of cavitation damage. In the present disclosure includes spot faces **340** with increased effectiveness without adversely impacting other performance parameters of the pressure exchanger such as mixing or efficiency.

(192) In some embodiments, pressure exchanger **300** may have a spot face **340** at the LPIN port without having a spot face **340** on LPOUT port.

(193) As a duct **322** carrying high pressure fluids approaches the LPIN and LPOUT ports, the rate of depressurization of the duct fluids can be controlled by incorporating a spot face **340** at the entrance of either LPIN port or at the entrance of LPOUT port or both. LPOUT port pressure is lower than LPIN port pressure due to the viscous and inertial losses in the ducts **322** of rotor **320**. In some embodiments (e.g., a typical SWRO plant), while using a pressure exchanger **300** as an energy recovery device, LPOUT pressure can be about half of LPIN pressure. As such, by locating the spot face **340** of a given geometry at LPIN entrance instead of LPOUT entrance, duct depressurization occurs with a lower pressure gradient (e.g., lower jetting velocities) and may result in lower cavitation potential, noise, and vibration levels.

(194) In some embodiments, pressure exchanger **300** includes a spot face **340** at LPIN port (e.g., instead of LPOUT port) and/or at HPOUT port (e.g., instead of HPIN port).

(195) As a duct **322** carrying low pressure fluids approaches the HP ports, the rate of pressurization of the duct fluids can be controlled by incorporating a spot face **340** at the entrance of either HPIN port or at the entrance of HPOUT port or both. HPOUT port pressure is lower than HPIN port pressure due to the viscous and inertial losses in the rotor ducts. As such, by locating the spot face **340** of a given geometry at HPOUT entrance instead of HPIN entrance, duct pressurization occurs with a lower pressure gradient (e.g., lower jetting velocities) and results in lower cavitation potential, noise, and vibration levels.

(196) In some embodiments, ducts **322** open to the spot face **340** at the duct inside diameter (ID) instead of at the duct outside diameter (OD). Fluid in the duct **322** undergoes rotary motion and, as such, the tangential component of the fluid velocity increases linearly with radius. In addition, the fluid pressure within the duct increases from duct ID to OD due to the centrifugal head caused by the centripetal acceleration of the fluid. When a duct carrying high pressure fluids approaches the spot face **340** at the entrance of a low pressure kidney, the depressurization occurs through jetting along the spot face **340** from duct ID instead of duct OD. This can be accomplished by adjusting the relative angle between the spot face **340** and the duct **322** at their approach. This can also be accomplished by curving the duct wall and/or the spot face **340** wall at their approach either towards or away from the duct **322**.

(197) In some embodiments, a chamfer at LPIN port (e.g., LPIN open) may change the included

angle of spot face **340** for different rows of ducts **322**. In some embodiments, the spot face **340** may conform to a duct **322**. In some embodiments, spot faces **340** may have a radial extent that is shorter than that of a duct **322**. Both the fluid tangential velocity and pressure in rotor ducts may increase with the radial coordinate. A way to prevent jetting and depressurization at the highest radius is by shortening the spot face **340** radial extent (e.g., spot face **340** with lower radial extent that is short than the radial extent of the duct **322**).

(198) In some embodiments, an end cover **310** forms split spot faces **340** (e.g., one per row of ducts or combined for a set of rows).

(199) A pressure exchanger **300** may include multiple rows of ducts **322** formed by the rotor **320**. Each of the rows may have a different duct geometry resulting in different volumes of fluids that are to be pressurized and/or depressurized. The fluid tangential velocity and pressure may change from one row to another. A spot face **340** tuned for each row of ducts **322** may be more effective at reducing noise, vibration, and cavitation than a single spot face **340** covering all of the rows. In some embodiments, a single spot face **340** is used that has a varied geometry along the radius (e.g., for the ramp angle).

(200) In some embodiments, the duct volume may be optimized (e.g., one or more rows of trapezoidal ducts). In some embodiments, ducts in adjacent concentric rows may be staggered.

(201) Splitting the rotor ducts into multiple rows instead of a single row while keeping the total rotor duct volume substantially the same may result in ducts of smaller size. By staggering the duct rows such that each row of the ducts approaches the kidneys at a different instant allows for depressurization and/or pressurization of a single duct in a single row at a time. This allows for shallower spot face **340** ramp and a reduced pressure gradient across the duct wall.

(202) Staggering the duct rows may result in a more uniform flow and pressures (e.g., minimize individual duct volume) resulting in reduced vibration levels.

(203) In some embodiments, various spot face **340** geometries may be used and may be combined with one or more pre-pressurization or depressurization holes.

(204) In some embodiments, one or more of spot face **340** angular extent, radial extent, and/or ramp may be optimized for different fluids.

(205) The depth, radial, and angular extents of the spot face **340** may depend on the pressure exchanger geometry (e.g., duct size, duct wall thickness, etc.), pressure exchanger flow rate, HP/LP pressure ratio, rotor RPM, type of fluid, etc. In some embodiments, the higher the compressibility of the fluid is, the higher the spot face **340** volume is to be to provide pressurization and/or depressurization of the rotor duct.

(206) The present disclosure may be used to reduce cavitation potential, noise, and/or vibration levels. The present disclosure may use spot faces **340** in addition to reducing RPM, reducing the number of pressure exchange cycles, including pre-pressurization and/or depressurization holes, and/or the like.

(207) The present disclosure may reduce noise generation, vibration levels, and cavitation potential in energy recovery applications using a pressure exchanger.

(208) The present disclosure may be used for use of pressure exchangers for energy recovery such as in SWRO, trans-critical CO₂ refrigeration/heat pumps, industrial waste water etc.

(209) The preceding description sets forth numerous specific details, such as examples of specific systems, components, methods, and so forth, in order to provide a good understanding of several embodiments of the present disclosure. It will be apparent to one skilled in the art, however, that at least some embodiments of the present disclosure may be practiced without these specific details. In other instances, well-known components or methods are not described in detail or are presented in simple block diagram format in order to avoid unnecessarily obscuring the present disclosure. Thus, the specific details set forth are merely exemplary. Particular implementations may vary from these exemplary details and still be contemplated to be within the scope of the present disclosure.

(210) Reference throughout this specification to “one embodiment” or “an embodiment” means

that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of the phrase “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. In addition, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” When the term “about,” “substantially,” or “approximately” is used herein, this is intended to mean that the nominal value presented is precise within $\pm 10\%$. Also, the terms “first,” “second,” “third,” “fourth,” etc. as used herein are meant as labels to distinguish among different elements and can not necessarily have an ordinal meaning according to their numerical designation.

(211) The terms “over,” “under,” “between,” “disposed on,” and “on” as used herein refer to a relative position of one material layer or component with respect to other layers or components. For example, one layer disposed on, over, or under another layer may be directly in contact with the other layer or may have one or more intervening layers. Moreover, one layer disposed between two layers may be directly in contact with the two layers or may have one or more intervening layers. Similarly, unless explicitly stated otherwise, one feature disposed between two features may be in direct contact with the adjacent features or may have one or more intervening layers.

(212) Although the operations of the methods herein are shown and described in a particular order, the order of the operations of each method may be altered so that certain operations may be performed in an inverse order or so that certain operation may be performed, at least in part, concurrently with other operations. In another embodiment, instructions or sub-operations of distinct operations may be in an intermittent and/or alternating manner. In one embodiment, multiple metal bonding operations are performed as a single step.

(213) It is to be understood that the above description is intended to be illustrative, and not restrictive. Many other embodiments will be apparent to those of skill in the art upon reading and understanding the above description. The scope of the disclosure should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which each claim is entitled.

Claims

1. A pressure exchanger comprising: a rotor configured to rotate to exchange pressure between a first fluid at a first pressure and a second fluid at a second pressure, wherein the rotor forms ducts that are routed from a first distal end of the rotor to a second distal end of the rotor, wherein the rotor forms chamfers on trailing edge rotor duct walls without forming chamfers on leading edge rotor duct walls; a first end cover disposed at the first distal end of the rotor, wherein the first end cover forms a high pressure in (HPIN) port configured to provide the first fluid at the first pressure into the ducts, and wherein the first end cover forms a low pressure out (LPOUT) port configured to receive the first fluid from the ducts at a third pressure that is lower than the first pressure; and a second end cover disposed at the second distal end of the rotor, wherein the second end cover forms a low pressure in (LPIN) port configured to provide the second fluid at the second pressure into the ducts and forms a high pressure out (HPOUT) port configured to receive the second fluid from the ducts at a fourth pressure that is higher than the second pressure.

2. The pressure exchanger of claim 1, wherein the second end cover forms a first spot face proximate the HPOUT port.

3. The pressure exchanger of claim 2, wherein the pressure exchanger forms a second spot face proximate the LPIN port of the second end cover without forming a spot face proximate the LPOUT port of the first end cover to depressurize a high pressure (HP) duct of the rotor at the LPIN port.

4. The pressure exchanger of claim 1, wherein the first end cover forms the HPIN port without forming a spot face proximate the HPIN port.

5. The pressure exchanger of claim 2, wherein the second end cover forms a chamfer at the LPIN port that is configured to change a corresponding angle of the first spot face for each concentric row of the ducts of the rotor.
6. The pressure exchanger of claim 5, wherein the chamfer at the LPIN port is configured to change an angle for each duct to be exposed to the first spot face to change an amount of pressurization of different rows of ducts.
7. The pressure exchanger of claim 2, wherein the first spot face has a radial extent that is shorter than a duct radial extent of a corresponding duct formed by the rotor to reduce a diameter at which a rotor duct is to open to an end cover port to reduce tangential velocity and cavitation potential.
8. The pressure exchanger of claim 2, wherein the first spot face comprises a first recess associated with a first concentric row of the ducts and a second recess associated with a second concentric row of the ducts.
9. The pressure exchanger of claim 1, wherein the rotor forms the ducts in concentric rows that are staggered.
10. The pressure exchanger of claim 1, wherein a duct opening of the rotor to a corresponding end cover spot face is at an inside duct diameter by adjusting an angle of a spot face edge or curving a spot face wall respective to the duct opening.
11. A pressure exchanger comprising: a rotor configured to rotate to exchange pressure between a first fluid at a first pressure and a second fluid at a second pressure, wherein the rotor forms ducts that are routed from a first distal end of the rotor to a second distal end of the rotor, wherein the rotor forms the ducts in at least two concentric rows comprising a first concentric row of the ducts, a second concentric row of the ducts, and a third concentric row of the ducts; and a first end cover disposed at the first distal end of the rotor, wherein the first end cover forms a first port, a second port, and a split spot face proximate the first port, wherein the split spot face comprises a first recess associated with the first concentric row of the ducts and a second recess associated with the second concentric row of the ducts and the third concentric row of the ducts.
12. The pressure exchanger of claim 11, wherein corresponding ducts in the first concentric row and the second concentric row are staggered.
13. The pressure exchanger of claim 11, wherein the split spot face has a radial extent that is shorter than a duct radial extent of a corresponding duct formed by the rotor.
14. The pressure exchanger of claim 11, wherein the pressure exchanger forms the split spot face proximate a low pressure in (LPIN) port without forming a spot face proximate a low pressure out (LPOUT) port to depressurize a high pressure (HP) duct of the rotor at the LPIN port.
15. The pressure exchanger of claim 11, wherein the rotor forms chamfers on trailing edge rotor duct walls.
16. A pressure exchanger comprising: a rotor configured to rotate to exchange pressure between a first fluid at a first pressure and a second fluid at a second pressure, wherein the rotor forms ducts that are routed from a first distal end of the rotor to a second distal end of the rotor, wherein the rotor forms the ducts in at least two concentric rows comprising a first concentric row of the ducts and a second concentric row of the ducts; and a first end cover disposed at the first distal end of the rotor, wherein the first end cover forms a first port, a second port, and a spot face, wherein the first end cover forms a chamfer at the first port, wherein the chamfer changes a corresponding angle of the spot face for each concentric row of the ducts of the rotor, and wherein the chamfer at the first port is configured to change an angle for each duct to be exposed to the spot face to change an amount of pressurization of different rows of ducts.
17. The pressure exchanger of claim 16, wherein the rotor forms the ducts in concentric rows that are staggered, wherein the spot face has a radial extent that is shorter than that of a corresponding duct to cause fluid tangential velocity and pressure in the corresponding duct to increase with a radial coordinate.
18. The pressure exchanger of claim 16, wherein the pressure exchanger forms the spot face

proximate a low pressure in (LPIN) port without forming a spot face proximate a low pressure out (LPOUT) port to depressurize a high pressure (HP) duct of the rotor at the LPIN port.

19. The pressure exchanger of claim 16, wherein the rotor forms chamfers on trailing edge rotor duct walls.
